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FINAL TERM ACTIVITY 3 _PYTHON WORD CLOUD

Documentation

```
# Install wordcloud if needed
!pip install wordcloud

# Imports
from wordcloud import WordCloud, STOPWORDS
import pandas as pd
import matplotlib.pyplot as plt

# URL for colormaps
# https://matplotlib.org/stable/tutorials/colors/colormaps.html
```

```
Requirement already satisfied: wordcloud in /usr/local/lib/python3.13/dist-packages (1.9.6)
Requirement already satisfied: numpy>=1.19.3 in /usr/local/lib/python3.13/dist-packages (from wordcloud) (2.4.6)
Requirement already satisfied: pillow in /usr/local/lib/python3.13/dist-packages (from wordcloud) (11.3.0)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.13/dist-packages (from wordcloud) (3.10.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib->wordcloud) (1.4.0)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.13/dist-packages (from matplotlib->wordcloud) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib->wordcloud) (4.68.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib->wordcloud) (1.5.1)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib->wordcloud) (26.3)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib->wordcloud) (3.3.3)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.13/dist-packages (from matplotlib->wordcloud) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (from python-dateutil>=2.7->matplotlib->wordcloud) (1.17.0)
```

```
# Import Data
df = pd.read_csv('https://raw.githubusercontent.com/SuperDataworld/Python/main/REF%20TWEETS.csv')

# Unify Text from all weeks
text = df['Tweet'].str.cat(sep=' ')


#set the stopwords list
stopwords= set(STOPWORDS)
new_words = []
new_stopwords=stopwords.union(new_words)

# Size of Word Cloud
plt.rcParams["figure.figsize"] = (10,10)

# Make Wordcloud
wordcloud = WordCloud(max_font_size=50, max_words=100, background_color="white",stopwords=new_stopwords, colormap='flag').generate(text)

# Plot Wordcloud
plt.plot()
plt.imshow(wordcloud, interpolation="bilinear")
plt.axis("off")
plt.show()

# Unify Text from tweets of different sentiment
text = df[df['Sentiment'] == 'Negative']['Tweet'].str.cat(sep=' ')

wordcloud = WordCloud(max_font_size=50, max_words=50, background_color="white", stopwords=new_stopwords, colormap='flag').generate(text)
plt.plot()
plt.imshow(wordcloud, interpolation="bilinear")
plt.axis("off")
plt.show()
```



Code for First Output:

```
!pip install wordcloud


import pandas as pd

import numpy as np

import matplotlib.pyplot as plt


from wordcloud import WordCloud, STOPWORDS

from PIL import Image
```

```
from google.colab import files


# Import Data
df =
pd.read_csv('https://raw.githubusercontent.com/SuperDataworld/Python
/main/REF%20TWEETS.csv')


# Display the dataset
df.head()


# Check column names
print(df.columns)


uploaded = files.upload()


# Load the uploaded JRU logo
logo = Image.open('JRU_thumbnail.png').convert('RGB')


plt.figure(figsize=(8, 5))
plt.imshow(logo)
plt.axis('off')
plt.show()


# Convert logo into a NumPy array
logo_array = np.array(logo)


# Identify the gold-colored parts of the logo
gold_mask = (
    (logo_array[:, :, 0] > 150) &
    (logo_array[:, :, 1] > 100) &
    (logo_array[:, :, 2] < 130)
)
```

```
# WordCloud uses black areas as drawable areas
mask = np.where(gold_mask, 0, 255).astype(np.uint8)

# Display the mask
plt.figure(figsize=(8, 5))
plt.imshow(mask, cmap='gray')
plt.axis('off')
plt.show()

# JRU color palette
navy_blue = "#102F69"
gold_yellow = "#F5B400"

def jru_color_func(word, font_size, position, orientation,
                   random_state=None, **kwargs):
    return gold_yellow

# Generate WordCloud
wordcloud = WordCloud(
    width=1200,
    height=800,
    background_color=navy_blue,
    mask=mask,
    stopwords=STOPWORDS,
    color_func=jru_color_func,
    max_words=200,
    contour_width=0,
    random_state=42,
    collocations=False
).generate(text)

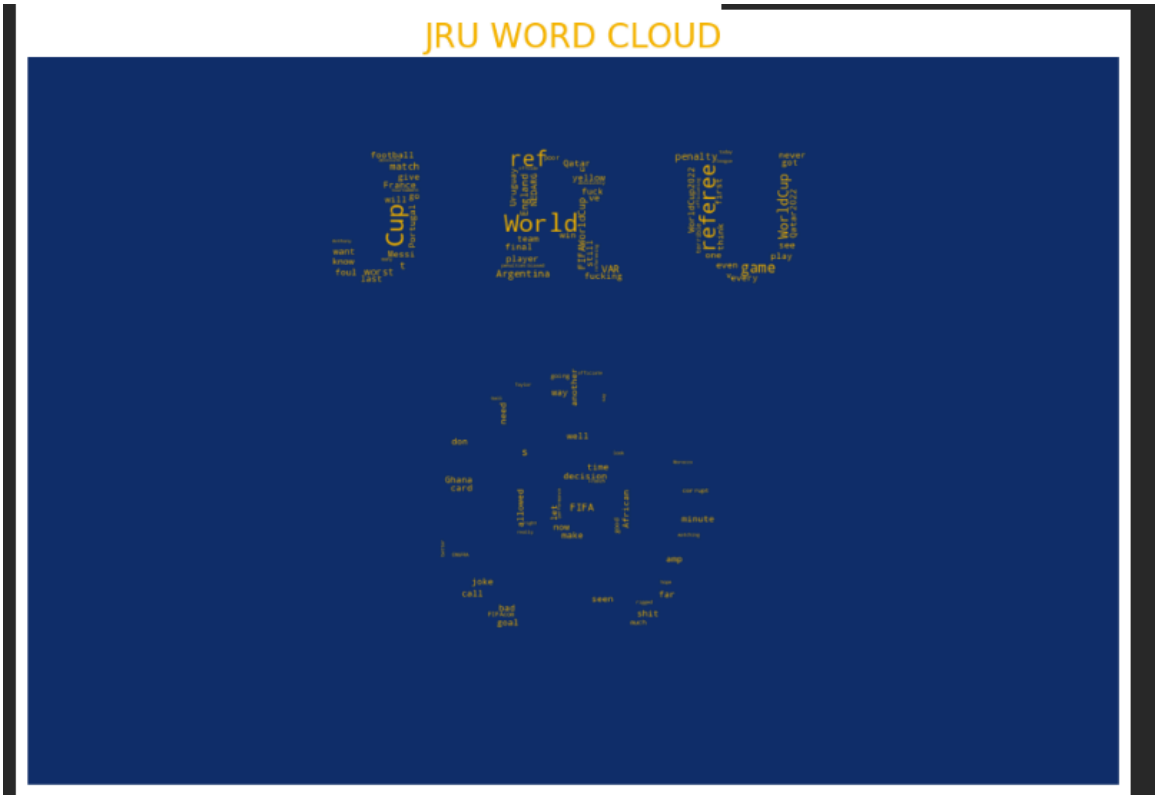
# Display WordCloud
```

```
plt.figure(figsize=(12, 8))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title("JRU WORD CLOUD", color=gold_yellow, fontsize=20)
plt.show()

# Save the generated WordCloud
wordcloud.to_file("JRU_WordCloud.png")

# Download the image
files.download("JRU_WordCloud.png")
```

First Output:



Code for Second Output:

```
!pip install wordcloud


import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from wordcloud import WordCloud, STOPWORDS
from PIL import Image
from google.colab import files
from io import BytesIO


# Load dataset
df =
pd.read_csv('https://raw.githubusercontent.com/SuperDataworld/Python
/main/REF%20TWEETS.csv')


# Combine tweet text
text = "
".join(df.select_dtypes(include='object').fillna('').astype(str).val
ues.flatten())


# Upload NBA logo
uploaded = files.upload()
logo =
Image.open(BytesIO(next(iter(uploaded.values())))).convert("RGB")


# Resize logo
logo.thumbnail((1200, 800))


# Create mask
logo_array = np.array(logo)


mask = np.where(
    np.any(logo_array < 240, axis=2),
    0,
```

```
    255
).astype(np.uint8)

# JRU color palette
navy_blue = "#102F69"
gold_yellow = "#F5B400"

# Gold word color
def jru_color_func(word, font_size, position, orientation,
                   random_state=None, **kwargs):
    return gold_yellow

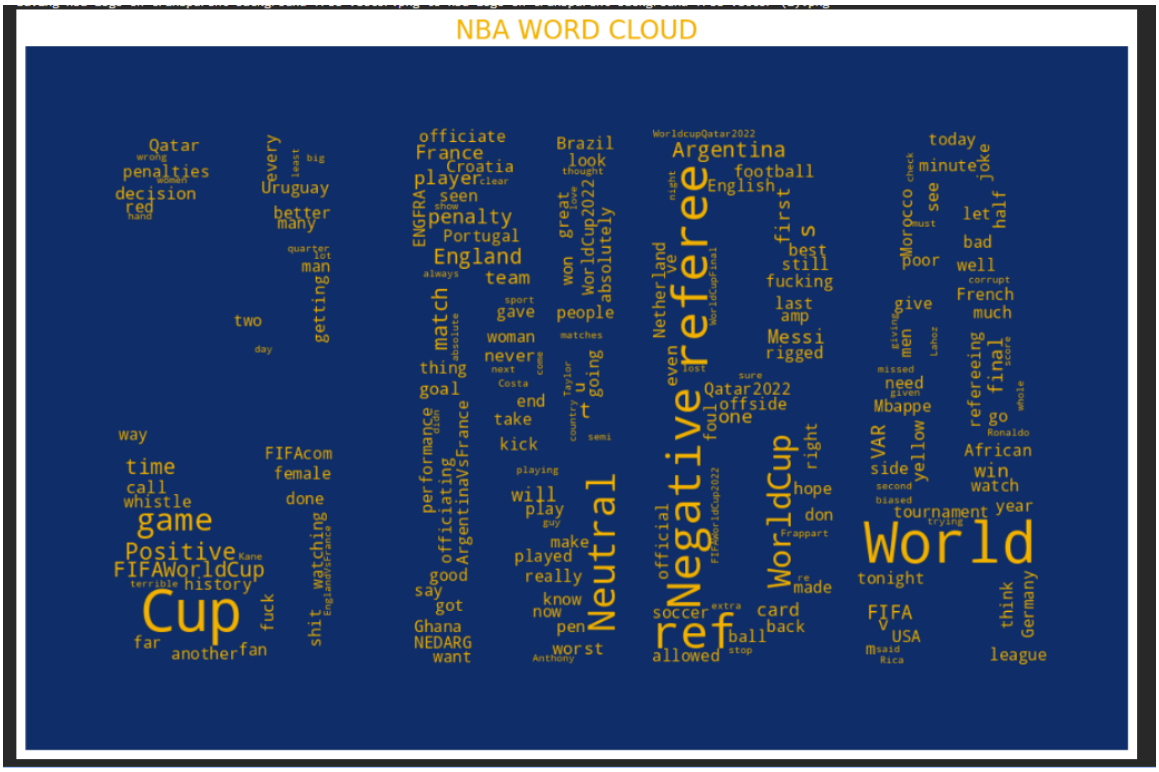
# Generate WordCloud
wordcloud = WordCloud(
    width=1200,
    height=800,
    background_color=navy_blue,
    mask=mask,
    stopwords=STOPWORDS,
    color_func=jru_color_func,
    max_words=200,
    contour_width=0,
    random_state=42,
    collocations=False,
    prefer_horizontal=0.8
).generate(text)

# Display final output only
plt.figure(figsize=(12, 8))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title("NBA WORD CLOUD", color=gold_yellow, fontsize=20)
```

```
plt.tight_layout(pad=0)

plt.show()
```

Second Output:



Code for Third Output:

```
!pip install wordcloud


import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import requests


from wordcloud import WordCloud, STOPWORDS

from PIL import Image

from io import BytesIO
```

```
# Load dataset
df =
pd.read_csv('https://raw.githubusercontent.com/SuperDataworld/Python
/main/REF%20TWEETS.csv')

text = " ".join(
    df.select_dtypes(include='object')
    .fillna('')
    .astype(str)
    .values.flatten()
)

# Load NCAA Philippines logo directly
logo_url =
"https://www.clipartkey.com/mpngs/m/263-2635556_ncaa-logo-philippine
s.png"

response = requests.get(logo_url)
response.raise_for_status()

logo = Image.open(BytesIO(response.content)).convert("RGBA")

# Resize logo
logo.thumbnail((1200, 800))

# Create mask using transparent and white background
logo_array = np.array(logo)

mask = np.where(
    (logo_array[:, :, 3] > 0) &
    (np.any(logo_array[:, :, :3] < 240, axis=2)),
    0,
```

```
    255
).astype(np.uint8)

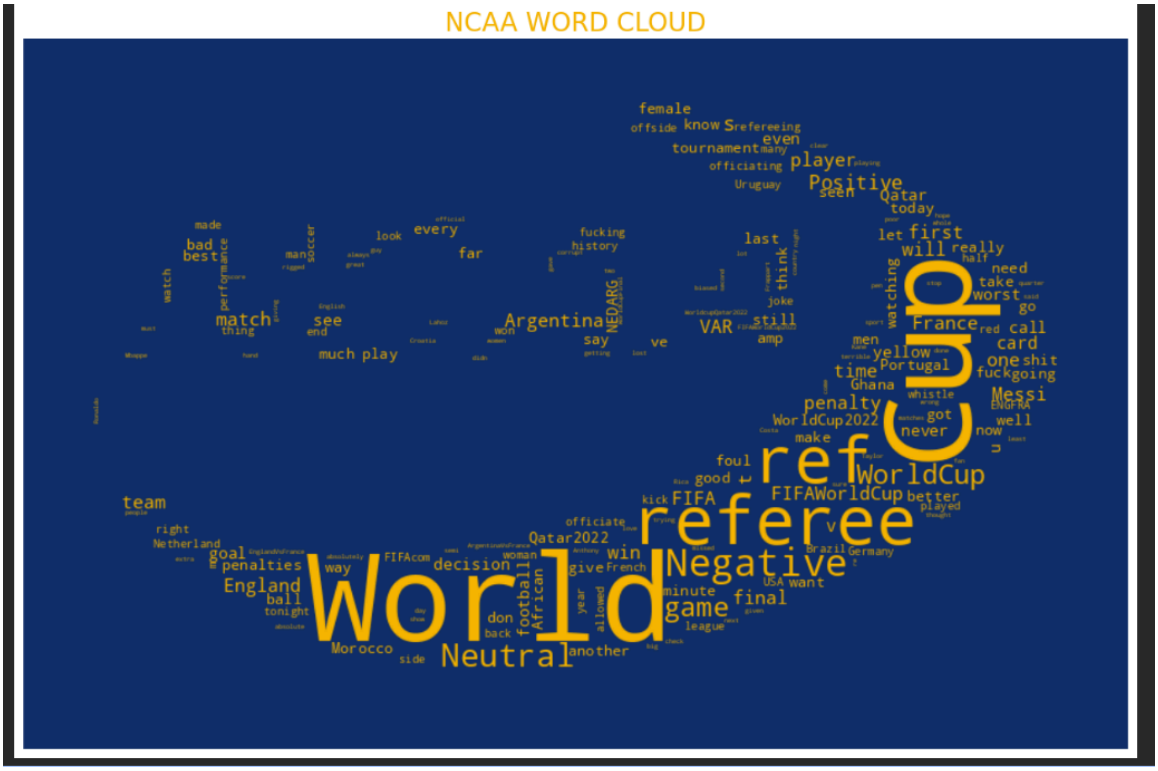
# NCAA color palette
navy_blue = "#102F69"
gold_yellow = "#F5B400"

def color_func(word, font_size, position, orientation,
               random_state=None, **kwargs):
    return gold_yellow

# Generate WordCloud
wordcloud = WordCloud(
    width=1200,
    height=800,
    background_color=navy_blue,
    mask=mask,
    stopwords=STOPWORDS,
    color_func=color_func,
    max_words=200,
    contour_width=0,
    random_state=42,
    collocations=False
).generate(text)

# Display final output only
plt.figure(figsize=(12, 8))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title("NCAA WORD CLOUD", color=gold_yellow, fontsize=20)
plt.tight_layout(pad=0)
plt.show()
```

Third Output:



Code for Fourth Output:

```
!pip install wordcloud


import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import requests


from wordcloud import WordCloud, STOPWORDS

from PIL import Image

from io import BytesIO


# Load dataset
```

```
df =
pd.read_csv('https://raw.githubusercontent.com/SuperDataworld/Python
/main/REF%20TWEETS.csv')


text = " ".join(
    df.select_dtypes(include='object')
    .fillna('')
    .astype(str)
    .values.flatten()
)


# Load JRU Bombers logo
logo_url =
"https://images.seeklogo.com/logo-png/39/1/jru-bombers-logo-png_seek
logo-394361.png"


response = requests.get(logo_url)
response.raise_for_status()


logo = Image.open(BytesIO(response.content)).convert("RGBA")
logo.thumbnail((1200, 800))


# Create logo mask
logo_array = np.array(logo)


mask = np.where(
    (logo_array[:, :, 3] > 0) &
    (np.any(logo_array[:, :, :3] < 240, axis=2)),
    0,
    255
).astype(np.uint8)


# JRU colors
```

```
navy_blue = "#102F69"
gold_yellow = "#F5B400"


def color_func(word, font_size, position, orientation,
               random_state=None, **kwargs):
    return gold_yellow


# Generate WordCloud
wordcloud = WordCloud(
    width=1200,
    height=800,
    background_color=navy_blue,
    mask=mask,
    stopwords=STOPWORDS,
    color_func=color_func,
    max_words=200,
    contour_width=0,
    random_state=42,
    collocations=False
).generate(text)


# Display final output only
plt.figure(figsize=(12, 8))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title("JRU BOMBERS WORD CLOUD",
          color=gold_yellow, fontsize=20)
plt.tight_layout(pad=0)
plt.show()
```

Fourth Output:

